Jenny Vivian

Academic affiliation: Adjunct fellow at UniSC Personal website: Jenny Vivian Australia, Austria (resident), Italy (citizen)

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Research Interests

Forest ecology, forest restoration, community ecology, ecosystem dynamics, Machine Learning, eDNA

Education

University of the Sunshine Coast Nov. 2023 – Aug. 2026
PhD in Forestry Studies
Thesis: "The effect of reforestation using *Acacia mangium* on above- and belowground ecosystem properties"
Principal Supervisor: Professor D. Lee; co-supervisors: Professor R. L. Chazdon, Professor A. Shapcott, Dr. A. A. Catling

University of Parma Sept. 2021 – July 2023
Master's Degree in Ecology and Ethology for Nature Conservation
Thesis: "The Colombian socio-ecological system: dynamics of deforestation between conflicts and conservation"
Final grade: 110/110 cum laude and Honorable Mention
Supervisor: Professor A. Bodini, co-supervisor: Dr. N. Clerici

University of Ca' Foscari, Venice Sept. 2018 – July 2021
Bachelor's degree in Environmental Sciences
Thesis "Abiotic stresses mitigation by NK fertilizer on seeds of ancient cereals"
Final grade: 110/110 cum laude
Supervisor: Professor G. Buffa; co-supervisor: Dr. M. Roscini

Work Experience

Sequench Ltd Nov. 2024 – March 2025
Internship on environmental DNA sequencing, bioinformatics, and biostatistics analyses

University of the Sunshine Coast Feb. 2024 – July 2024
Teaching assistant for "Discovering Science"

University of Parma Oct. 2022 – March 2023
Teaching assistant for "Biological laboratory techniques (Ecology module)"

Selected publications

- **Vivian, J.**, Chazdon, R. L., Shapcott, A., & Lee, D. J. (2025).
  Toward a deeper understanding of recovery trajectories in forest ecosystem restoration: a machine learning approach.
  Restoration Ecology, e70483
- **Vivian, J.**, Chazdon, R. L., Catling, A. A., Shapcott, A., Herbohn, J., & Lee, D. J. (2026).
  Acacia mangium monocultures can catalyse the recovery of the tree community and aboveground carbon stock in the Philippines.
  Forest Ecology and Management, 601, 123357
- **Vivian, J.**, Scriver, M., Zaiko, A., Shapcott, A., Chazdon, R. L., Catling, A. A., & Lee, D. J. (2026).
  Recovery of soil microbiota in naturally regenerating *Acacia mangium* ecosystems.
  PeerJ 14:e21048
- **Vivian, J.**, Chazdon, R. L., Catling, A. A., & Lee, D. J. (2025).
  Early evidences of links between soil microbes and forest restoration through multiple ecosystem metrics.
  Frontiers in Forests and Global Change, 8, 1540513
- **Vivian, J.**, Shapcott, A., Chazdon, R. L., & Lee, D. J. (2026).
  Microbial rhizosphere and topsoil communities in young reforestation plantations.
  Frontiers in Microbiology, 17, 1915742.
- Catling, A. A., Derisso, V. D., Gamalo, L. E. D., Herbohn, J., McRae, C. L., Mooney, M. G., **Vivian, J.**, & Putz, F. E. (2026).
  Logging legacies in the Australian tropics: 56-years of skid trail recovery.
  Forest Ecology and Management, 619, 124075
- Bodini, A., **Vivian, J.**, Vargas, J., Clerici, N., & Scotti, M. (2024).
  The web of conflict-related interactions in Colombia: exploring causal linkages between ecological and social variables by the qualitative loop analysis.
  Philosophical Transactions B, 379(1909), 20230165

Interests and Competences

Languages	Italian (Mother tongue); English (TOEFL iBT, with a score of 108/120, corresponding to a C1; 22-10-2022)
Programming and software skills	R, QGIS, Python, LaTeX, Past 4.03, basics of "Weka"
Driving licenses	A2-B (Italy); C-RE (Australia)
Voluntary activity (May 2025 – May 2026)	High Degree by Research Student Representative of the Forest Research Institute (FRI) Leadership Group at the University of the Sunshine Coast
Voluntary activity (Dec. 2024 – Present)	Lead convenor of Special Interest Network "Integrated Forest Restoration" for the Ecological Society of Australia

Awards and Honors

• Grant "Quantum and Advanced Technologies Talent Building Program" for the development of research on merging quantum computing in ecology, from the Queensland Government	20,000 AUD	Oct. 2025
• Culture Champion Award for "outstanding contributions helping to build and nurture the caring, supportive culture in the workplace"	50 AUD	Aug. 2025
• Prize for the development of my research (from 1st place in the 3-minute thesis competition at Forest Research Institute of the University of the Sunshine Coast	500 AUD	June 2025
• Travel bursary for Australian & New Zealand Environmental DNA (eDNA) Conference (Wellington)	2300 AUD	Feb. 2025
• Partial travel funding for Society of Ecological Restoration conference (Denver)	200 USD	Oct. 2025
• Travel Award from the Ecological Society of Australia, for ESA conference (Melbourne)	150 AUD	Dec. 2024
• Travel Award from the Theo Murphy Initiative (Australia) administered by the Australian Academy of Science, Ecological Society of Australia, Plant-Soil Ecology Special Interest Network, to attend the ESA conference and workshops (Melbourne)	1000 AUD	Dec. 2024
• Scholarship for Master's degree (2023) final grade, from the local authority	500 EUR	Jan. 2024
• Grant to participate in the Blended Intensive Programme "Towards a Sustainable Water Management" at the University of Extremadura, financed by ERASMUS+ SMS Short Mobility Grant, managed by the University of Parma in accordance with ERASMUS+ National Agency	Total cover	June 2023
• Grant to participate in the "International Summer School-Pollutions in Coastal Areas" at the University of Gdansk, from the project itself, financed under the Knowledge Education Development Operational Program of the EU Commission	Total cover	Sept. 2022
• Scholarship of merit, 2nd place, year 2018-2019, from Ca' Foscari University of Venice	1000 EUR	March 2020

Conferences and Presentations

• UNESCO Chair on Sustainable Management of Conservation Areas: presentation "How is *Acacia mangium* affecting forest restoration in the Philippines, Jenny?"	Online, Oct. 2026
• European conference on ecological restoration (SERE2025; Brest, August 2026): presentation "Assessment of forest restoration projects: efficiency of cutting-edge techniques and comprehensive analysis" and "Optimized spatial planning in forest restoration via quantum algorithms: a proof of concept for viable plans"	Brest, Aug. 2026
• Forestry Australia Conference: presentation "Trajectories of forest ecosystem recovery with *Acacia mangium* in the Philippines: what made it a successful story"	Adelaide, Oct. 2025
• World conference on ecological restoration (SER2025): presentation results "Can *Acacia mangium* plantations develop properties of remnant forests? A Machine Learning perspective"	Denver, Sept. 2025
• International Conference on Conservation Biology (ICCB): presentation "Recovery of tree communities and above-ground carbon stock in *Acacia mangium* reforestation sites"	Brisbane, June 2025
• Australian & New Zealand Environmental DNA (eDNA) Conference: presentation "Evaluating soil fungi and bacteria in *Acacia mangium* plantations and remnant forests in the Philippines using environmental DNA"	Wellington, Feb. 2025
• Ecological Society Conference (ESA): poster presentation of early results	Melbourne, Dec. 2024
• University of the Sunshine Coast Showcase: poster presentation "The effect of reforestation using *Acacia mangium* on above- and below-ground ecosystem properties"	Sippy Downs, Sept. 2024
• Airborne eDNA workshop (attended online): presentation "Air DNA analysis and Machine Learning: a promising collaboration"	Canberra, June 2024
• Climate Exp0 (Online): presentation "Sustainable agriculture approaches"	May 2021

Relevant Accomplishments

Admitted to the PhD Program in Ecology at the University of Zurich/ETH Zurich in 2023. No PhD positions were available at the time.